Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Hounds of Singularity
b. Team leader name: Harshwardhan Tripathi
c. Problem Statement: Media, Content & Digital Experiences

Team member: Chennika

Brief about the idea

The rail keeps media identity separate from what a caption claims.

The web workspace and X extension are its first API clients. Both can retrieve stored evidence; new claims can start durable Google Cloud jobs.

Gemini assesses fetched source quotations and critiques conclusions. Uncertain evidence stays unresolved; media origin remains unknown.

Opportunities

Reusable infrastructure: an investigation should be available wherever the same claim appears, with its context and limitations intact.

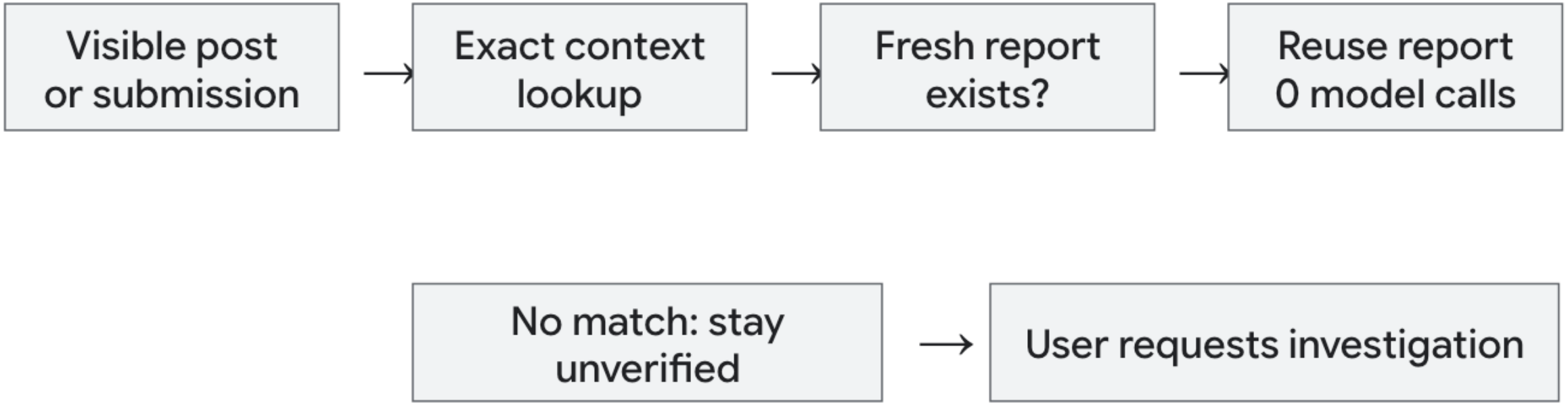
Current clients: the web workspace and X extension.

Future consumers could include social apps, Android/Gemini experiences and agents through an MCP adapter. These are proposals, not existing integrations or partnerships.

List of features offered by the solution

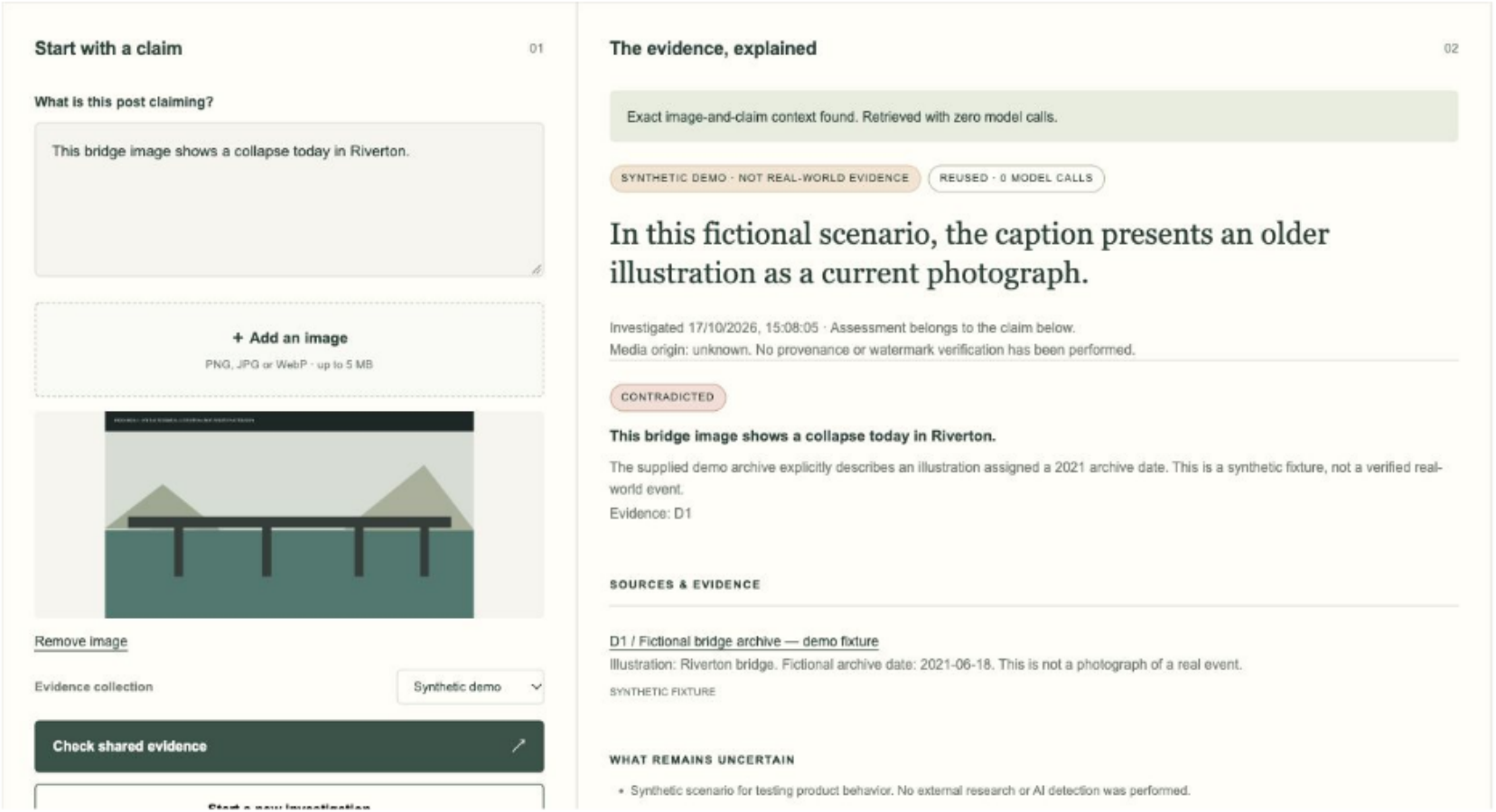
- Shared lookup, investigation-job and report APIs
- Exact image, caption and publication-context identity
- Findings with validated source quotations and context checks
- Persistent jobs and isolated reviewer access with reserved allowance
- Public recorded examples; explicit failure and uncertainty states

Process flow diagram or Use-case diagram

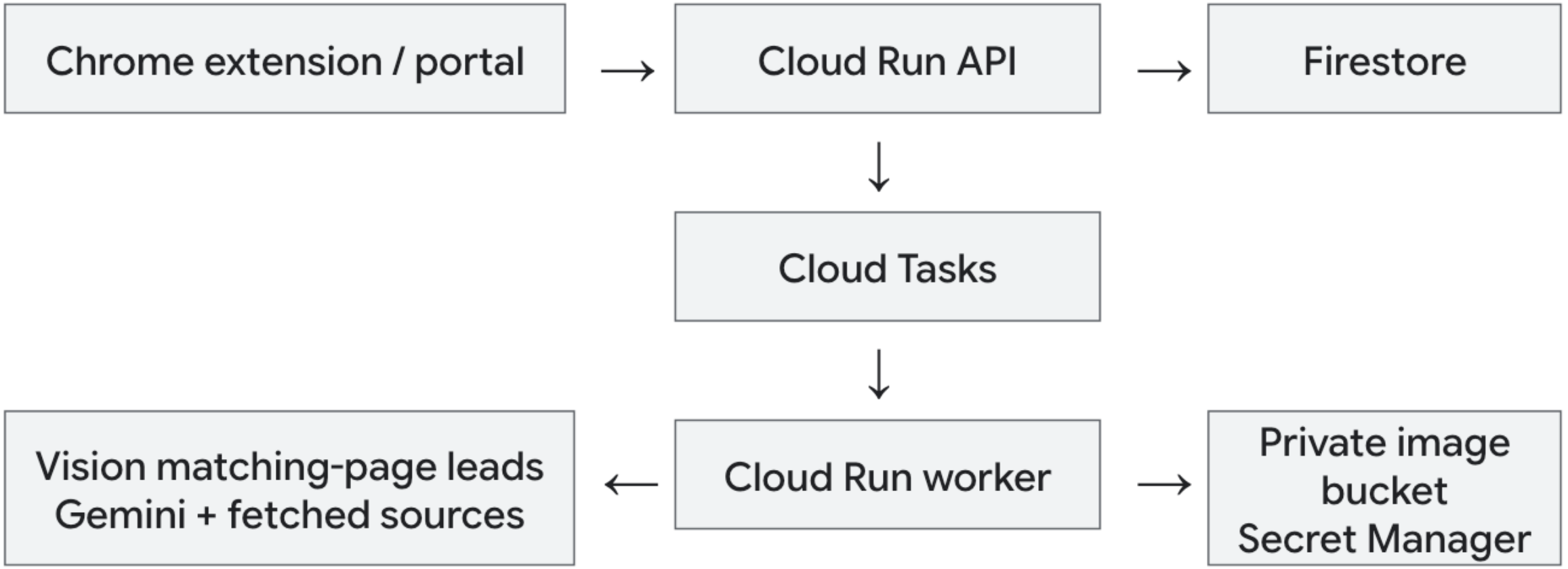


A matching image does not answer a new caption.

Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Mumbai resources; Gemini global endpoint. Attached service identities.

Technologies to be used in the solution

Gemini on Vertex AI: search research, structured assessment and critique
Cloud Vision Web Detection: image matching-page leads
Cloud Firestore: reports, jobs and transactional admission limits
Cloud Run + Cloud Tasks: application and durable work delivery
Cloud Storage + Secret Manager: temporary images and credentials
FastAPI, Python, Chrome Manifest V3, HTML/CSS/JavaScript

Estimated implementation cost (optional)

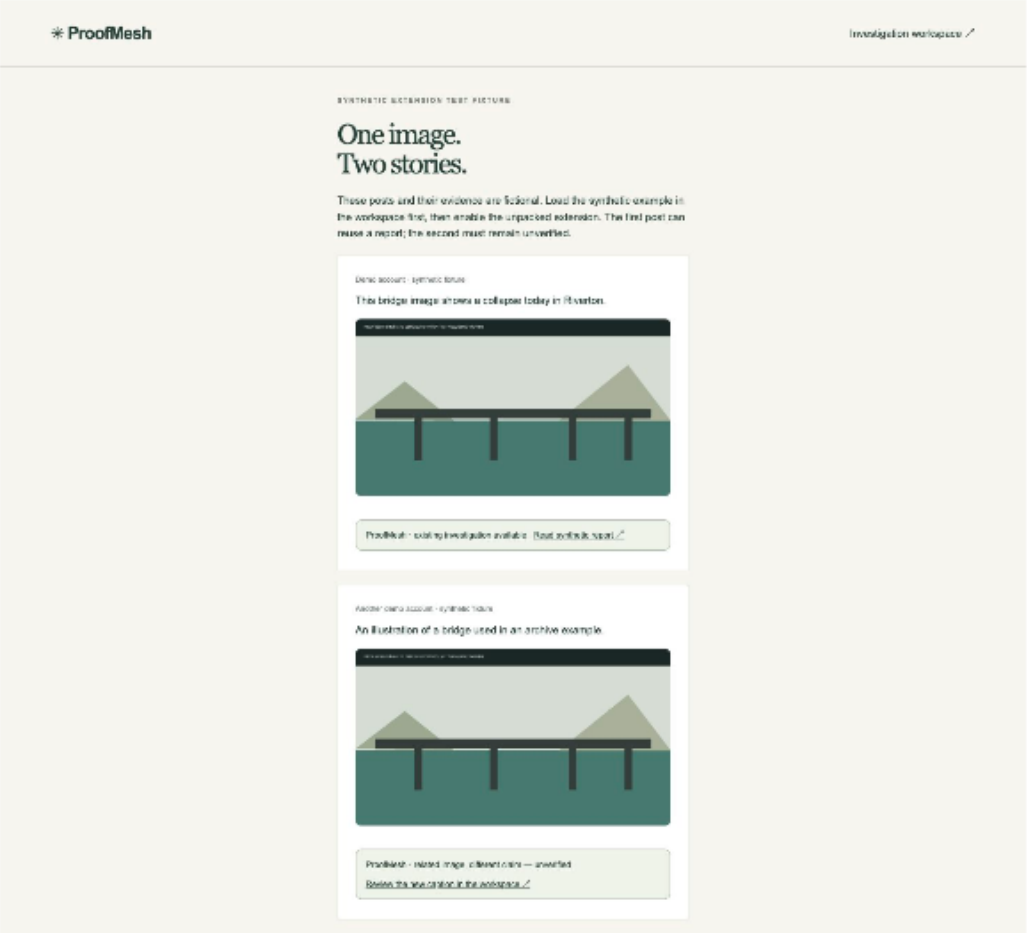
Approved trial-credit budget: US$5 total for bounded work

Twelve-case development evaluation: US$0.994 working estimate, including conservative failed-attempt reserves. Not a billing receipt; some infrastructure costs excluded.

Controls: zero minimum / one maximum instance; four admissions per UTC day and twelve across the live window, with six reserved for reviewers.

No billing upgrade. Alerts and workload limits are not an absolute dollar cap.

Snapshots of the prototype





Prototype Performance report/Benchmarking

Twelve prespecified development cases: 11 expected outcomes, nine conclusive findings and three abstentions. A known false flag assertion stayed unresolved after quotation validation failed.

First pass: nine completed; three provider 429 failures. Explicit follow-ups completed all three. Twelve repeats: zero model calls.

47 Python + 27 JavaScript checks pass. Latest hosted reviewer inference hit 429 twice and failed safely. This is not representative detection accuracy.

Additional Details/Future Development (if any)

Next rail slice: a stable API contract, small SDK and MCP adapter, with explicit permission for paid investigations. Not implemented yet.

A pilot needs tenant isolation, revocable access, report revisions, contribution review and capacity monitoring.

Current limits: single-image X adapter; no Instagram adapter, video analysis, watermark detector or independent fact verification.

Project links

Public repository
https://github.com/htg2704/proofmesh

Demo video (3 minutes 20 seconds)
https://proofmesh-2god3tua3a-el.a.run.app/assets/demo.mp4

Working prototype
https://proofmesh-2god3tua3a-el.a.run.app/

Evidence and evaluation notes

Primary references: NASA Earthrise, ISRO Chandrayaan-3, National Capital Authority and the European Union flag page.
Earthrise image credit: NASA/Bill Anders; no endorsement.

Public samples are actual v0.4 recorded outputs. Earlier model errors and provider failures remain documented. Viewing a recorded case makes no inference calls.

Method, case labels and limits: docs/EVALUATION.md and evaluation/results-v04.json in the repository.

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Thank you

